

Connectivity Hitting in the Random Graph Process: Exact Finite Laws and the Gumbel Window

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Abstract

Reveal the $K = \binom{n}{2}$ edges of the complete labeled graph in uniform random order, and let τ_{conn} be the first connected prefix. Decomposition by the component containing vertex 1 gives an exact recurrence for every connected-graph count, hence the complete finite CDF, PMF, tail, support, and all moments of τ_{conn} . At $m = \lfloor (n/2)(\log n + c) \rfloor$, isolated-vertex factorial moments and a uniform spanning-tree bound excluding every other component prove $2\tau_{\text{conn}}/n - \log n \Rightarrow$ the standard Gumbel law.

1 The monotone process and connected counts

Let $e_1, \dots, e_K$ be a uniformly random permutation of the edges of K_n , and put $G_m = (\{1, \dots, n\}, \{e_1, \dots, e_m\})$. Thus G_m is uniform over the $\binom{K}{m}$ labeled graphs with m edges. Edge addition is monotone, and the connected graphs form an absorbing upper set. We take $\tau_{\text{conn}} = \min\{m : G_m \text{ is connected}\}$, with $\tau_{\text{conn}} = 0$ when $n = 1$.

Write $C(n, m)$ for the number of connected labeled graphs on n vertices with m edges. Set $C(1, 0) = 1$, and interpret both $C(s, j)$ and binomial coefficients as zero outside their natural ranges.

Theorem 1 (All finite connectivity-hitting laws). *For $n \geq 2$, $K = \binom{n}{2}$, and every integer m ,*

$$C(n, m) = \binom{K}{m} - \sum_{s=1}^{n-1} \binom{n-1}{s-1} \sum_j C(s, j) \binom{\binom{n-s}{2}}{m-j}. \quad (1)$$

Moreover $C(n, m) = 0$ for $m < n - 1$,

$$C(n, n-1) = n^{n-2}, \quad C(n, K) = 1. \quad (2)$$

For $0 \leq m \leq K$, with $F_n(-1) = 0$,

$$F_n(m) := \mathbb{P}(\tau_{\text{conn}} \leq m) = \frac{C(n, m)}{\binom{K}{m}}, \quad (3)$$

$$\mathbb{P}(\tau_{\text{conn}} = m) = F_n(m) - F_n(m-1), \quad (4)$$

$$\mathbb{P}(\tau_{\text{conn}} > m) = 1 - F_n(m). \quad (5)$$

For every integer $r \geq 1$,

$$\mathbb{E}[\tau_{\text{conn}}^r] = \sum_{m=0}^{K-1} ((m+1)^r - m^r) \{1 - F_n(m)\}. \quad (6)$$

Finally, for $n \geq 2$,

$$n-1 \leq \tau_{\text{conn}} \leq \binom{n-1}{2} + 1. \quad (7)$$

Proof. If a graph is disconnected, let s be the size of the component containing vertex 1. Choose its other $s-1$ labels, choose its connected j -edge subgraph, and place every remaining edge among the other $n-s$ vertices. This decomposition is unique and subtracting it from all m -edge graphs gives (1). A connected graph needs at least $n-1$ edges; at equality it is a tree, counted by the Prüfer-code value n^{n-2} . The complete graph is unique, proving (2).

The first m positions of a uniform edge permutation form a uniform m -subset, which proves (3). Monotonicity gives (4)–(5). For every nonnegative integer-valued T , $T^r = \sum_{m=0}^{T-1} ((m+1)^r - m^r)$; expectation proves (6). The smallest connected graph is a tree. A disconnected graph has at most $\binom{n-1}{2}$ edges, since convexity maximizes the sum of within-component edge capacities at component sizes $n-1$ and 1. This proves (7). $\square$

2 The connectivity window

For fixed $c \in \mathbb{R}$, let

$$m_n(c) = \left\lfloor \frac{n}{2}(\log n + c) \right\rfloor. \quad (8)$$

This lies in $[0, K]$ for all sufficiently large n . Let I_n be the number of isolated vertices in $G(n, m_n(c))$.

Lemma 1 (Poisson isolated vertices). *I_n converges in law to $\text{Poisson}(e^{-c})$.*

Proof. For fixed $r \geq 1$, an ordered r -tuple is isolated exactly when every chosen edge lies among the other $n - r$ vertices. Hence, with $K = \binom{n}{2}$,

$$\mathbb{E}(I_n)_{r\downarrow} = (n)_{r\downarrow} \frac{\binom{n-r}{m_n(c)}}{\binom{K}{m_n(c)}}. \quad (9)$$

Put $D_r = K - \binom{n-r}{2} = rn - r(r+1)/2$ and $m = m_n(c)$. Since $m = O(n \log n)$,

$$\begin{aligned} \log \frac{\binom{K-D_r}{m}}{\binom{K}{m}} &= \sum_{j=0}^{m-1} \log \left(1 - \frac{D_r}{K-j} \right) \\ &= -D_r \frac{m}{K} + O \left(\frac{D_r m^2}{K^2} + \frac{D_r^2 m}{K^2} \right) = -r(\log n + c) + o(1). \end{aligned}$$

Together with $(n)_{r\downarrow} \sim n^r$, formula (9) tends to e^{-rc} . These are all factorial moments of $\text{Poisson}(e^{-c})$; the factorial-moment criterion proves the claim. $\square$

Isolated vertices are the only asymptotically relevant obstruction. The following proof retains the without-replacement law rather than silently switching to $G(n, p)$.

Lemma 2 (No other component survives). *At $m = m_n(c)$,*

$$\mathbb{P}\{G(n, m) \text{ is disconnected and } I_n = 0\} \longrightarrow 0.$$

Proof. Let X_s count components with exactly s vertices. If a fixed s -set is a component, some one of its s^{s-2} spanning trees is present and all $b = s(n-s)$ crossing edges are absent. For a fixed tree with $a = s-1$ edges, sampling m edges without replacement gives

$$\begin{aligned} \mathbb{P}\{A \subseteq G_m, B \cap G_m = \emptyset\} &= \frac{(m)_{a\downarrow} (K-m)_{b\downarrow}}{(K)_{a+b\downarrow}} \\ &\leq \left(\frac{m}{K}\right)^a \exp \left\{ -\frac{b(m-a)}{K-a} \right\}. \end{aligned} \quad (10)$$

Consequently

$$\mathbb{E}X_s \leq \binom{n}{s} s^{s-2} \left(\frac{m}{K}\right)^{s-1} \exp \left\{ -\frac{s(n-s)(m-s+1)}{K-s+1} \right\}. \quad (11)$$

Here $m/K \asymp (\log n)/n$. Uniformly for $2 \leq s \leq n/\log n$, the exponent in (11) is at least $s(\log n - O_c(1))$; using $\binom{n}{s} \leq (en/s)^s$ and summing gives

$$\sum_{2 \leq s \leq n/\log n} \mathbb{E}X_s \leq \sum_{s \geq 2} n \left(\frac{C_c \log n}{n} \right)^s = o(1).$$

For $n/\log n \leq s \leq n/2$, one has $n-s \geq n/2$ and $m-s+1 \geq m/2$ for large n . The exponent is then at least $s \log n/8$, while the remaining factors are at most $n(C \log n)^s$. Thus

$$\sum_{n/\log n \leq s \leq n/2} \mathbb{E}X_s \leq \sum_{s \geq n/\log n} n \left(\frac{C \log n}{n^{1/8}} \right)^s = o(1).$$

If a graph is disconnected and has no isolated vertex, one of its components has size between 2 and $\lfloor n/2 \rfloor$. Markov's inequality and the two sums prove the lemma. $\square$

Theorem 2 (Gumbel connectivity hitting law). *For every real c ,*

$$\lim_{n \rightarrow \infty} \mathbb{P} \left\{ \frac{2\tau_{\text{conn}}}{n} - \log n \leq c \right\} = \exp\{-e^{-c}\}. \quad (12)$$

Proof. Lemmas 1–2 give $\mathbb{P}\{G(n, m_n(c)) \text{ connected}\} = \mathbb{P}(I_n = 0) + o(1) \rightarrow \exp\{-e^{-c}\}$. Because τ_{conn} is integer, the event on the left of (12) is exactly $\{\tau_{\text{conn}} \leq m_n(c)\}$; equation (3) finishes the proof. $\square$

Source ownership and AI-use statement

Erdős and Rényi [1] established the random-graph evolution and connectivity threshold. We claim no priority for that result. The present package supplies a self-contained frozen-model derivation, exact finite hitting atlas, and auditable boundary ledger. AI tools assisted symbolic checks, hostile mutation design, and manuscript preparation; all proof obligations, cutoffs, and nonclaims are explicit.

References

- [1] P. Erdős and A. Rényi, “On the evolution of random graphs,” *Publications of the Mathematical Institute of the Hungarian Academy of Sciences* 5 (1960), 17–61.